Hasibe Nida Akdoğan

Software Developer | Python & Data Analytics

Düzce / Istanbul, Türkiye (open to remote and relocation) | hasibenidaakdogan@gmail.com

linkedin.com/in/nidaakdogan | github.com/nidakd | nidaakdogan.vercel.app

PROFESSIONAL SUMMARY

Software Developer with a B.Sc. in Mathematics (English) and an associate degree in Computer Programming, building data-driven applications with Python, SQL and FastAPI. Designed and built an event-driven, machine-learning-assisted decision engine end to end, and has delivered projects ranging from an AI legal research assistant to full-stack web applications. Combines mathematical, analytical thinking with technical team leadership.

TECHNICAL SKILLS

- **Programming Languages:** Python, SQL, C#, JavaScript, TypeScript, HTML, CSS
- **Data & Analysis:** pandas, NumPy, Matplotlib, Seaborn, Jupyter Notebook, Data Preprocessing, Data Annotation, Data Visualization, Web Data Extraction (Python, regex), Backtesting, Microsoft Excel (Advanced)
- **Databases:** PostgreSQL, SQLAlchemy, Alembic, Database Design, Query Optimization
- **Frameworks & Libraries:** FastAPI, Streamlit, ASP.NET Core Web API, Entity Framework Core, React, Tailwind CSS, Pydantic, pytest, VTK, Vedo
- **Tools & Platforms:** Git, GitHub, Docker, VS Code, Visual Studio, pgAdmin, Postman, Swagger, Supabase, Gemini API (LLM integration)
- **Concepts:** REST APIs, WebSockets, Event-Driven and Hexagonal Architecture, Unit Testing, Responsive Web Design, 3D Mesh Processing, Mathematical Modeling, Statistics, Probability, Linear Algebra, Analytical Problem Solving

EXPERIENCE

Mentor | GDG on Campus Abant Izzet Baysal University | Bolu, Türkiye | 09/2026 – Present

- Mentor community members and the technical team on software development projects across different technologies.

Software & Technical Team Lead | GDG on Campus Abant Izzet Baysal University | Bolu, Türkiye | 09/2025 – 09/2026

- Led a software and technical team of 5+ members, coordinating project development lifecycles and mentoring members.
- Co-organized 48 workshops, study groups and coding events that reached 1,500+ participants.
- Designed and taught a Python 101 workshop as instructor and event coordinator, covering variables, data types, lists, tuples and control flow with Google Colab materials.

Web Design Intern | Serbay Interactive | 01/2026 – 02/2026

- Developed CMS-based websites on Kod8 CMS, structuring layouts with HTML and CSS and managing content and site optimization through management panels.
- Ensured responsive, cross-device compatibility and consistent color and visual design; applied mockup adjustments and UI refinements for brand consistency.
- Sourced digital assets and edited and optimized vector graphics with Inkscape; batch-converted and resized images with Converseen and organized project files; created and updated QR menus for restaurant clients.
- Wrote Python scripts (standard library, regex) to extract company data from web page sources into deduplicated, Excel-ready CSV files and to validate city and district records in SQL database dumps against a reference list.

Software Development Intern | Armada Yazılım (Bilişim Vadisi) | 08/2025 – 09/2025

- Labeled about 1,000 3D intraoral scans for AI model training, color-coding each tooth according to defined annotation criteria in a Python-based viewer (VTK, Vedo).
- Processed 3D mesh data into training-ready datasets, gaining hands-on experience with mesh structures.
- Collaborated through Git-based version control and regular team meetings, preparing progress summaries.

Site Accountant (Volunteer) | Residential Site Management | 06/2022 – Present

- Manage the financial operations of a residential complex, communicating regularly with 60 residents about dues and payments.
- Prepare and analyze monthly financial reports on income, expenses, utility billing and outstanding balances using advanced Excel.
- Hold bank account authorization; record incoming payments and reconcile bank transactions with accounting records.

PROJECTS

NDA Trading – Algorithmic Decision Engine | Python, FastAPI, PostgreSQL, NumPy, React, TypeScript | 06/2026 – Present | Private repository

- Built an asynchronous, event-driven decision core for crypto futures (Binance USDT-M) that ingests live market data, scores signals across multiple strategies, filters them with machine learning and executes orders under risk rules; hexagonal (ports and adapters) architecture: about 13,900 lines of Python across 218 commits.
- Implemented a NumPy-vectorized indicator engine (RSI, EMA, ATR, ADX, Bollinger Bands), multiple trading strategies, a risk service with position sizing and drawdown guards, and a paper-trading simulator for risk-free testing.
- Developed a backtesting pipeline and an online logistic-regression signal filter in NumPy; tuning volatility filters cut trade count, and with it commission costs, by 80% in backtests.
- Exposed a JWT-secured FastAPI REST and WebSocket API with Prometheus metrics, a React and TypeScript dashboard, PostgreSQL storage via SQLAlchemy and Alembic, Telegram notifications and 187 automated pytest tests.

Site Accounting App | Python, Streamlit, PostgreSQL | github.com/nidakd/site-accounting-app

- Designed and built an accounting system for residential site management to increase financial transparency and prevent misuse of funds.
- Tracks dues, fuel costs and carry-over debts per building block, with FIFO-based payment allocation, automated receipt generation and CSV import on a modular architecture.

EmsalAI – AI Legal Research Assistant | GDG Hackathon '26, Ostim Technical University | 04/2026

- Built in a 3-person team during a 24-hour hackathon: an AI assistant that finds relevant statutes and precedent decisions for legal cases.
- Contributed to both the backend (asynchronous FastAPI with Google Gemini 2.5 Flash Lite and Server-Sent Events for real-time streaming) and the frontend (React, Tailwind CSS, Supabase).

Product Stock Management | Capstone project, Web Build Program | ASP.NET Core Web API (.NET 10), Entity Framework Core, React, PostgreSQL | github.com/nidakd/product-stock-api

- Built a full-stack inventory application with CRUD operations, search, validation, logging and automatic database setup, exposing a Swagger-documented REST API.

EDUCATION

B.Sc. in Mathematics (English) | Abant Izzet Baysal University | Bolu, Türkiye

Associate Degree in Computer Programming | Anadolu University | 2026

CERTIFICATIONS & TRAINING

- Web Build Program (6-week full-stack training: React, ASP.NET Core Web API, PostgreSQL) | YOBODOBO | 04/2026
- GitHub Workshop | Management Information Society | 03/2026
- SQL / PostgreSQL Bootcamp (Data Science Bootcamp SQL Program) | TurkStudentCo | 05/2025
- SQL Excellence Bootcamp | Techcareer.net | 11/2024
- Python & AI Fundamentals (Introduction to Python; First Step into AI; Akbank Python Bootcamp) | Global AI Hub | 02/2024 – 03/2024

LEADERSHIP & VOLUNTEERING

- Mathematics Mentor | NASA International Space Apps Challenge 2025, TED Bolu College | 10/2025
- Instructor | Python 101 Workshop, GDG on Campus Abant Izzet Baysal University | 11/2025

LANGUAGES

Turkish (Native) | English (B2) | French (A2)